

Transitions in Glomerular Disease

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Keywords

Glomerular disease · Health care transition · Adolescence · Chronic kidney disease · Pediatric

Abstract

Background: The transition from pediatric to adult care is a complex and challenging issue for increasing numbers of adolescents and young adults (AYA). Despite increasing attention to the transition of AYA with chronic medical conditions, there remain difficulties which vary by local resources, disease processes, and individual factors. There are also poorly understood factors regarding transition involving rare diseases, including glomerular disease. Specialties and organizations are working to evaluate and enhance the transition process; however, the variability in resources limits formation of a true consensus. **Summary:** AYA with glomerular disease, broadly as chronic kidney disease and specifically with respect to diagnoses such as lupus nephritis, are vulnerable during the health care transition (HCT). Though young, many of these patients have severe disease with the loss of kidney function over their disease course and risk significant complications around the time of transition. Many facets make HCT challenging related to personal, caregiver, provider, and system factors. Despite difficulties, some groups have found success, largely in dedicated clinics designed to promote successful transition from pediatric to adult models of care.

Society guidelines focus on a structured, intentional process providing continuity, with or without a dedicated transition clinic. Preparation, communication across the transition, and support for patients and providers with additional staff are all recommended aspects of transition from the International Society of Nephrology and International Pediatric Nephrology Association. Additionally, transition toolkits are available from several sources which can aid in developing a structured transition program for patients with glomerular disease. **Key Messages:** (1) There are many challenges in transitioning AYA patients with glomerular disease from pediatric to adult care. (2) HCT is a vulnerable period with increased risk of poor outcomes. (3) A successful HCT may be promoted by developing a structured process and/or program focused on preparation, communication, and support.

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Introduction

Adolescence and young adulthood present challenges for patients with chronic diseases, particularly glomerular diseases. In this period, adolescents and young adults (AYA) face important life transitions but additionally have the demands of managing a chronic disease. Childhood-onset glomerular diseases are commonly lifelong diseases with times of remission and

relapse. Glomerular diseases are a group of diseases characterized by inflammation and destruction or other damage to the glomeruli resulting in the need for ongoing treatment and transition from the pediatric care model to an adult care model. There are personal challenges such as struggling to connect with peers, managing demanding school and/or work schedules along with medical appointments and treatments, and navigating complexities of the health care system through changes in doctors or clinics as they approach adulthood and an adult model of care. The Society of Adolescent Health and Medicine defines health care transition (HCT) as the process of moving from a pediatric to an adult model of health care with or without a transfer to a new health care provider [1]. Transition therefore represents the entire process including preparation and education, while transfer specifically indicates the time the patient establishes with an adult provider or clinic.

The transition process represents a period with increased risk for poor health care outcomes in patients with chronic disease, but specific support for HCT may mitigate some of these risks. To achieve a successful transition, Six Core Elements were identified by the American Academy of Pediatrics, American Academy of Family Physicians, and the American College of Physicians [2]. Many societies and specialties have begun to identify particular challenges and specific interventions to improve the transition process, and some groups such as the International Society of Nephrology (ISN) and the International Pediatric Nephrology Association (IPNA) have published consensus statements regarding transition [3]. We will review the available literature on the scope of the challenge of HCT for individuals with chronic glomerular disease and strategies to mitigate risks in this vulnerable time.

Challenges of Transition

Though there is little dedicated literature regarding HCT in glomerular disease, it is generally known to be a difficult time particularly for those with chronic illnesses which continue from childhood into adulthood. Many facets of their disease, treatment, access to care, and long-term outcomes may require dedicated attention and discussion (Table 1). Suboptimal HCT may have lasting ill effects not only on the disease outcomes and access to care for AYA but also on their quality of life, education, and employment opportunities [4]. A review of adult survivors of childhood onset nephrotic syndrome

highlighted a lack of published literature regarding outcomes despite indicating 15% (range 8–33%) continued to have relapses of nephrotic syndrome in adulthood [5]. This review also details the limited data concerning long-term outcomes in cardiovascular health, growth, and fertility which are theorized to be affected by the disease process or treatments received.

There is also literature describing specific challenges and outcomes in patients who have lupus nephritis as a facet of childhood-onset systemic lupus erythematosus (cSLE). Retrospective studies assessing outcomes of patients with cSLE showed prolonged gaps in care between the last pediatric and first adult rheumatology visit, decreased medication adherence, and increased acute health care utilization [6–9]. One cSLE cohort identified through private insurance claims demonstrated 25% of patients were lost to follow-up after their last pediatric rheumatology and/or nephrology visits [7]. In another cohort of 50 patients with cSLE, mostly from moderate- to high-income households (92%), a mean time of 9 months elapsed between the last pediatric rheumatology visit and the first adult visit [6]. About 28% of these patients had active renal disease in the 3 years following transfer to adult care, and 2 out of the 14 patients with active renal disease went on to develop End-Stage Renal Disease (ESRD). Additionally, young adults (aged 19 to 39 years) with SLE have high rates of cardiovascular disease which leads to a 14-fold increased mortality compared to the general population [10].

In addition to these specific outcomes, we additionally can infer likely outcomes from similar disease states such as those involving immunosuppression or chronic kidney disease (CKD). For example, the Chronic Kidney Disease in Children (CKiD) study suggested that an increased number of medications increased the rate of nonadherence and identified a higher medication burden in those with glomerular disease compared to other etiologies of CKD [11]. In kidney transplant recipients, who also require immunosuppressive medication, graft loss may be the highest between the ages of 17 and 24 years regardless of the recipient's age at the time of transplantation [12] and particularly after transition to an adult unit [13]. A review of nonadherence in kidney transplant recipients including survey-based and some immunosuppression level monitoring studies further noted that nonadherence is much higher in AYA than in younger children or older adults [14]. A study in China suggested that over 50% of adolescent patients with kidney disease had a low level of transition readiness, particularly in those requiring immunosuppression [15]. In that cohort, living in a more rural area, paying out of

Table 1. Important topics for discussion and education in transition of patients with glomerular disease

Topic	Examples
<i>Disease-specific</i>	
Potential genetic contribution	Nephrotic syndrome: may impact response to treatment, risk of relapse, or posttransplant recurrence
Expected natural history of disease	Nephrotic syndrome: most children will have at least one relapse but most will not continue to have relapses into adulthood
Cosmetic effects and quality of life in active disease	SLE: rashes, hair loss, and edema
Cosmetic effects of treatments	Steroids: acne, weight gain, striae
Acute and chronic infectious risks of immunosuppression	Prophylaxis against <i>Pneumocystis carinii</i> may be appropriate for patients receiving high-dose steroids, rituximab, and cyclophosphamide among others
<i>Transition and access</i>	
Research	Increased access to clinical trials as an adult
New or alternate medications and monitoring/treatment guidelines as an adult	ANCA: avacopan approved in adults as an alternative to steroid use for adjunct therapy
Legal standing	Potential need for guardianship
Health insurance coverage	Loss of parent's insurance
<i>Long-term considerations</i>	
Cancer risks of immunosuppression	Mycophenolate: skin cancers, risk of blood dyscrasia in prolonged use; Cyclophosphamide: bladder cancer
Cardiovascular disease health	Potential for early and/or life-long antihypertensive or lipid-lowering medication
Reproductive health and family planning	Teratogenicity of medications; impact of pregnancy on autoimmunity and kidney function
Psychosocial health	School interruption for appointments, flares or relapses, and/or treatments
SLE, systemic lupus erythematosus.	

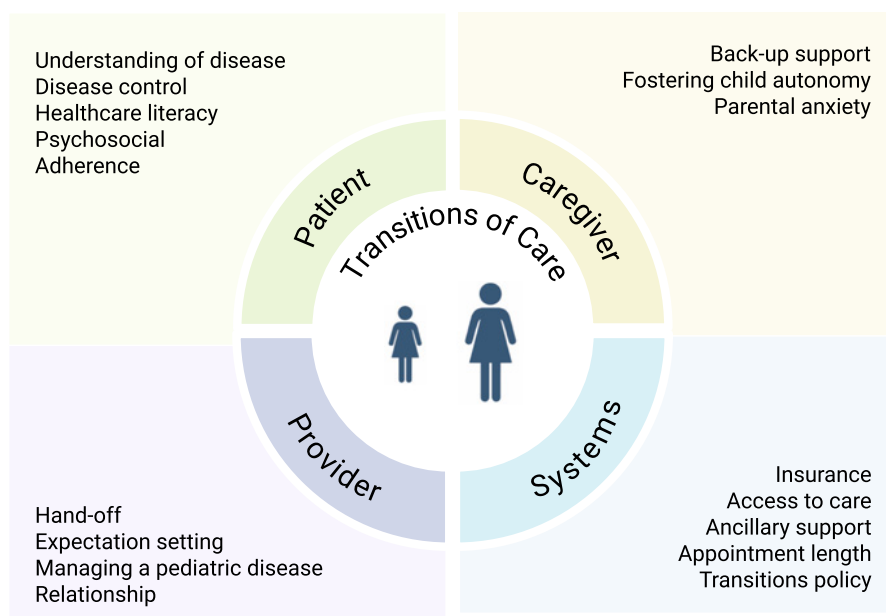
pocket rather than through insurance, and a lower household income were all associated with higher levels of transition readiness. It is clear that factors related to the patient, their family or caregivers, the providers, and the health system (shown in Fig. 1) contribute to challenges throughout the transition process and may have different impacts for different individuals.

Patient Factors

Many patient factors relate to challenges in physical and psychosocial development through adolescence and young adulthood. When facing chronic health conditions, such as glomerular disease, they must also continue appointments or treatments which may make them feel different and disconnected from their peers. These young patients may also struggle with feeling out of place in adult units or clinics when seeing older patients there

[12]. They are expected to learn new skills for self-management, and through transition, they often face changes to new health care providers or new physical locations providing their medical care. Developing self-management skills often requires support from the patient's caregiver and provider, requiring a balance of advancing autonomy with supervision [1]. These additional challenges contribute to an increased risk of nonadherence with their treatment plan, leading to increased risk for morbidity and mortality. Other causes of nonadherence may include side effects of medications. For example, steroids are commonly used in the treatment of glomerular diseases but may cause acne, weight gain including the development of striae, increased anxiety, and poor sleep, and all of these effects may have a negative psychosocial impact [16–18] which may create added complexity in the transition process for AYA with

Fig. 1. Top left: patient factors are understanding of disease, disease control, health care literacy, psychosocial, and adherence. Top right: caregiver factors are back-up support, fostering child autonomy, and parental anxiety. Bottom left: provider factors are hand-off between pediatric and adult care teams, expectation setting, appointment length, managing a pediatric disease, and therapeutic relationship. Bottom right: system factors are insurance, access to care, ancillary support, and a formal transition policy. Created in BioRender. Kerski, M. (2026) <https://www.biorender.com/bd7xz2m>.



CKD. Another potential side effect is an increased risk of infection from immunosuppression routinely used to treat glomerular diseases, though in some cases such as nephrotic syndrome, the condition itself also significantly increases the risk of infection [19].

Caregiver Factors

One study assessing transition readiness predictors identified that a sense of family cohesion and unity impacted patient readiness [20]. Parental anxiety in the transition process, particularly regarding differences in recommended treatments, also poses a challenge to a successful transition [21]. Indeed, clinicians have reported that parent's or caregiver's difficulty in "letting go" of pediatric locations and staff is a source of significant difficulty in the transition process [22]. However, some parenting styles and cultures may instead be protective during transition and may be leveraged to improve transition readiness [11].

Provider Factors

Health care providers, both pediatric and adult, also experience challenges with the transition process. Pediatric providers may be reluctant to transfer their patients, with one survey of post-transition young adults suggesting "earlier" as the way to improve their transition process [23]. Pediatric patients and providers find it challenging to identify a receiving adult provider [24]. Providing the receiving provider with a detailed medical and psychosocial summary may help ease the transition

period [24]. Adult nephrologists in one survey perceived the care of AYA as more difficult than of older patients citing the need for further medical information relevant to treating childhood-onset renal disease and for more psychosocial knowledge and support [21].

Disease etiologies, presentations, and management may vary between age at presentation and pediatric versus adult guidelines, and it is important for patients, caregivers, and providers to be aware of the differences. For example, patients with cSLE are more likely to have severe manifestations such as lupus nephritis than patients who first develop SLE in adulthood. Further discordance exists in nephrotic syndrome, where children presenting with nephrotic syndrome between the ages of 1 and 12 years are treated empirically with steroids but adults presenting with nephrotic syndrome typically undergo diagnostic kidney biopsy to determine treatment [19, 25]. While steroid response is determined within a maximum of 6 weeks of high-dose treatment in children, steroid response may not be determined until after 16 weeks of high-dose treatment in adults whose diagnoses warrant initial steroid trial.

System Factors

Barriers to successful HCT may also relate to the health system itself, with adult-focused clinics often having less multidisciplinary support, fewer ancillary services, and variable access to financial support for patients or families [17]. Indeed, a systematic review of survey-based assessments of transition identified patient

and caregiver dissatisfaction with adult renal units for reasons such as less time with nurses and providers [12]. Access to treatments and research such as clinical trials may vary between pediatric and adult clinics as well, with less frequent inclusion of pediatric populations in trials despite evidence to suggest that overall disease outcomes such as the loss of glomerular filtration rate are similar across age groups [26]. Cost of care and health care insurance also play a role; a lack or the loss of insurance in AYA is one predictor of failure to transition [17], and young adults form the largest population without insurance in the USA [11]. Because of this, specific attention to social determinants of health such as “socioecological models” is recommended [1]. Access to a multi-professional team was identified as a major difference between pediatric and adult units [27]. In the USA, Title V case management services are available, though not universally used, to assist eligible patients, families, and providers through the transition period [4]. Some glomerular disease also warrants comanagement with multiple specialists, such as rheumatology with lupus nephritis, which adds to the complexity of the transition.

Access to care and to insurance affects patients, providers, and outcomes. A significant reported concern from providers and health systems involves the cost of a transition clinic or program [21]. The question of cost-effective HCT is further complicated by uncertainty about cost surrounding this period. Specific cSLE cohorts, for example, have suggested disparate relative costs with some studies showing a higher acute care utilization [9] and others showing lower health care utilization [28] surrounding the transition period. One cohort showed reduced medication use and rheumatology follow-up after transition or transfer in part due to difficulties in obtaining insurance; these patients reported overall lower health and social functioning post-transition [28].

There is no best practice established on a specific age for transfer to adult care; many groups utilize a range of ages for different stages in the transition process including age ranges for establishing adult care to allow for an individualized progress [11, 27]. It is generally agreed that transition should occur at a time of disease control without flare or crisis [3, 29]. The age for transfer or completion of transition may also be impacted by governmental factors, hospital policy, or the availability of specialists at a particular geographic location or institution [22]. For example, if there is a shortage of pediatric specialists, pediatric patients may exclusively present to adult providers [3], or the transfer of care may occur earlier based on local availability.

Reproductive Health

CKD, including CKD due to glomerular disease, may also increase gynecologic and obstetric complexity for patients, families, and providers which may require particular attention for AYA. CKD is known to be disruptive to the hypothalamic-pituitary-ovarian axis which may lead to abnormal menstrual cycles including anovulation or abnormal uterine bleeding [30]. Additionally, both CKD and some immunosuppressive medications are known to affect fertility [5, 30] and many immunosuppressive or renoprotective agents are known teratogens [10]. Patients and providers may also have concerns regarding potential genetic contributors to disease which may affect treatment choices as well as family planning. Genetic mutations, particularly in podocyte genes, may be present in children with steroid-resistant nephrotic syndrome and may guide therapeutic and prognostic discussions and choices [19]. It is frequently hypothesized that patients with cSLE may have a stronger genetic underpinning to their disease as well. However, only 11% of a large ancestrally and geographically diverse cohort had a rare variant in monogenic SLE risk genes; most had common risk variants seen across the age spectrum [31]. Dedicated reproductive and sexual health screening and counseling are appropriate for all AYA with glomerular diseases.

Tools of Transition

Ideally, a combined transition effort with participants from both the pediatric and adult sides is proposed with one option from the ISN/IPNA consensus being a single transition clinic wherein patients meet with staff from both departments [3]. Where this is not possible, they recommend either a dedicated team including adult nephrologist, nurse, and ancillary staff or a dedicated young adult clinic which incorporated both transition patients and newly presenting young adults. They additionally promote the use of group sessions or peer support and propose expected patient competencies for transfer. Listed tools for preparing for transition or assessing transition readiness include a “transition medical passport” and the TRxANSITION Index [32]. Medical passports have been studied for various settings and may be customized with different features such as the inclusion of specialist contact information and medications and, in the case of transition, may include a transition coordinator or team where available [33].

Table 2. Summary of tool kits or frameworks to guide transition

Resource	Description
Got Transition (gottransition.org) [35]	Tool kit guided around Six Core Elements: 1. Developing a transition policy 2. Tracking and monitoring 3. Assessments for transition readiness 4. Transition planning 5. Transfer of care to adult model 6. Transfer completion
RISE to transition from Raina et al. [36] (2018)	Four competencies to assess transition readiness in kidney transplant recipients: 1. Recognition – understanding disease history and health system 2. Insight – understanding of treatment adherence on short- and long-term health 3. Self-reliance – gaining independence in scheduling/attending appointments, refilling medications, identifying changes in health 4. Establishing health lifestyle
American College of Rheumatology Pediatric to Adult Care Transition [37]	Tool kits developed for transitioning SLE and JIA: • Transition policy resources • Transfer letter • Forms to assist with documenting disease history • TRAQ/Transition Readiness Assessment for Youth (Got Transition)

SLE, systemic lupus erythematosus; JIA, juvenile idiopathic arthritis; TRAQ, Transition Readiness Assessment Questionnaire.

Similarly, a group of cSLE patients surveyed after transition stressed the importance of access to their online electronic medical record, multimodal disease-specific education materials, and social workers [34]. Further highlighting the importance of multidisciplinary, this group also surveyed health care workers caring for post-transition cSLE patients who advised the importance of partnership with psychologists and social workers along with early planning and preparation prior to transfer.

Different groups have proposed additional transition tools for different ages and populations (Table 2). The most well-known tool kit is Got Transition (gottransition.org) [35] which includes a variety of resources for patients and providers. Their tool kit centers around the Six Core Elements which include the early development of a transition policy individualized for each child and family, the use of transition tracking, determining transition readiness, planning for transition, the transfer or integration to adult-focused care, and finally the transition completion [2]. A systematic review specifically focusing on transition of patients with kidney transplant divides these competencies differently into a “RISE to transition” framework focusing on Recognition, Insight, Self-Reliance, and Establish(ing) healthy choices [36].

Transition readiness assessments (Table 3) can help determine areas requiring focused education or intervention for an individual patient during the transition process. The provider-administered TRxANSITION Index [32] and self-administered or parent-completed STARx questionnaire [38] are tools established by the University of North Carolina to provide measurable progress throughout a transition process and help determine transition readiness (<https://www.med.unc.edu/transition/transition-tools/>) [39] and are specifically recommended by the ISN/IPNA consensus [3]. Other commonly used tools include the self-administered patient Transition Readiness Assessment Questionnaire (TRAQ) [40] the Readiness for Transition Questionnaire (RTQ) which has parallel surveys for patient, parent/guardian, and provider and was originally developed for kidney transplant patients [41, 42]. Implementation of these questionnaires can be another challenge. One group published results from a quality improvement project in which an institution-wide transition readiness assessment, modified from TRAQ, was automated to be sent annually to patients of ages 14 years and over [43]. After automation, the rate of completion of this questionnaire dramatically improved from 12% to 90% and the response rate was sustained over the next 2 years. The authors attributed success to

Table 3. Common transition readiness assessments

Assessment	Self-administered	Parent-/caregiver-administered	Provider-administered	Example of use in nephrology or rheumatology
TRxANSITION Index [25]			X	Fenton [20] (2015)
STARx Self-Management and Transition Readiness Questionnaire [30, 45]	X	x		Yang [44] (2025)
TRAQ Transition Readiness Assessment Questionnaire [32]	x			Haro [28] (2020), Argraves [43] (2024)
RTQ Readiness for Transition Questionnaire [33]	X	X	X	Marchak [42] (2015)
TRANSITION-Q [46]	x			Morisaki-Nakamura [47] (2022)

automation and a motivated Quality Improvement (QI) team with division and institution level support. The American College of Rheumatology (ACR) has developed dedicated toolkits (<https://rheumatology.org/pediatric-to-adult-rheumatology-care-transition>) [37] for the transition of patients with juvenile idiopathic arthritis and lupus which provide resources on a transition policy, transfer letter, medical summary specific to lupus, the TRAQ [40], and the Transition Readiness Assessment for Youth (TRAY) [35].

Guidelines and Recommendations

Since the management of patients in the transition age group has distinct challenges, several groups related to the care of patients with glomerular disease have developed guidelines which in part address these challenges (Table 4). The ISN/IPNA 2011 consensus statement focuses on several components of transition including preparation, communication during transition and transfer, timing of transfer, a transfer plan and process, and best practices for a transition or transfer clinic and other ways to promote continuity of care [3]. In addition, disease-specific guidelines from the ACR and IPNA recognize the importance of an organized HCT. The 2025 ACR guidelines on SLE specifically recommend “a planned transition to adult care to decrease risks of unscheduled health care utilization, care gaps, and disease flares” [48], and the 2024 ACR guidelines on lupus nephritis recommend that for children with cSLE , a “structured, intentional transition” from pediatric to

adult rheumatology and nephrology care is “indicated to avoid poor outcomes during this vulnerable period,” which is included in their list of adjunctive/non-immunologic treatments [49].

The IPNA clinical practice recommendations on the diagnosis and management of immunoglobulin A nephropathy and immunoglobulin A vasculitis have a section on transition with eight recommendations related to planning, staffing, types of support, and duration of support [50]. The authors note that they cannot grade the evidence due to this topic being an area where validating studies have not been performed (either they are not able to be performed or there is clear benefit). Some vasculitis guidelines, however, lack specific recommendations regarding transition, such as the 2021 ACR/Vasculitis Foundation Guideline for the management of Anti-Neutrophilic Cytoplasmic Autoantibody (ANCA) vasculitis [51] and the Kidney Disease Improving Global Outcomes (KDIGO) 2025 Clinical Practice Guideline for the Management of immunoglobulin A nephropathy and immunoglobulin A vasculitis [52].

A disease-specific guide to transition of care in rheumatology was published in 2024. In addition to laying out overarching principles in the care of adults who have grown up with a systemic autoimmune disease, the guide also provides specific recommendations related to particular diagnoses including cSLE [10]. They note the particularly severe phenotype of cSLE and complications related to bone density, reproductive health, cardiovascular disease, and increased HPV prevalence which all require routine screening. Vaccination against HPV is also recommended, and attention

Table 4. Society guidelines addressing transition in glomerular disease

Society	Year	Topic	Reference
ISN/IPNA	2011	Transition from adult to pediatric renal services	Watson [3] (2011)
JSN/ JSPN	2018	Transition of patients with childhood-onset CKD	Kubota [29] (2018)
IPNA	2024	Diagnosis and management of children with IgA nephropathy and IgA vasculitis	Vivarelli [50] (2025)
ACR	2024	Lupus nephritis clinical care	Sammaritano [49] (2024)
ACR	2025	clinical care	Sammaritano [48] (2025)

ISN, the International Society of Nephrology; IPNA, the International Pediatric Nephrology Association; JSN, the Japanese Society of Nephrology; JSPN, the Japanese Society for Pediatric Nephrology; CKD, chronic kidney diseases; ACR, American College of Rheumatology; SLE, systemic lupus erythematosus.

to cardiovascular disease risk such as hypertension and hyperlipidemia is encouraged. The authors provide detailed suggestions regarding communication between providers and the patient or a parent/caregiver to set the care team up for success in managing different expectations between pediatric and adult care and ways to address the patient fully including supporting mental health. These guides are detailed and disease-specific, though not endorsed by a particular medical society.

Finally, some guidelines do not specifically address HCT but do discuss broad topics which may be important for many patients with glomerular disease and which may be of particular interest to AYA. The 2021 KDIGO Clinical Practice Guideline for the Management of Glomerular Diseases addresses edema, infection risk and vaccination, and cardiovascular concerns such as hypertension and hyperlipidemia [25]. The 2025 KDIGO Clinical Practice Guideline for childhood nephrotic syndrome recommends genetic testing for all patients with steroid resistance as well as those with syndromic features or family history of steroid resistant nephrotic syndrome.

Successful Implementation of Tools and Guidelines

Transition readiness is an important feature of mitigating the risks to patients undergoing HCT. By applying the TRxANSITION scale, a previous version of the TRxANSITION Index, to assess transition readiness in a cohort of nephrology patients with strong glomerular disease representation, researchers found that transition readiness was associated with reduced emergency room visits and increased medication adherence [20]. By utilizing the tools and recommendations to facilitate effective transitions, some groups have shown additional successful impacts. The

preferred intervention, a transition clinic or paired pediatric and adult clinics, has been attempted by many different groups. One early example is the integrated pediatric-young adult transition clinic for the transition of kidney transplant recipients at Oxford (UK) which resulted in improved graft survival [53]. A single center in Italy implemented a multidisciplinary transition program based on the RISE to transition framework. They found that the knowledge and satisfaction of patients and families were improved and reported a zero dropout rate after 2 years [54]. In the Kidney Adolescent and Young Adult Clinic (KAYAC) in South Africa, over 70 percent of included patients were listed as having glomerular disease. KAYAC found improved outcomes including reduction in the number of patients lost to follow-up and in mortality [45]. In Toronto, authors found success by utilizing a paired SickKids Transition Clinic (STC) in the pediatric space and the Adult Vasculitis Clinic [46] or the Young Adult SLE (YASLE) clinic in the adult space [55]. The STC included both pediatric and adult providers allowing for overlap of care during the transition process prior to completing transfer of care.

Short of building a specific transition clinic, other aspects of care have been identified as factors leading to success. For example, a group in Montreal found that identifying a single adult nephrologist and establishing a dedicated triage process and transition nurse improved follow-up [56]. In a post-transition study of semi-structured interviews of cSLE patients and health care professionals, both patients and health care team members stressed the importance of health literacy and patient activation, that is, the patient taking on a more active role in their care [34].

As many of these skills are evaluated in scoring tools such as the TRxANSITION Index or TRAQ, these assessments may be valuable for monitoring progression to transition readiness even though there is limited longitudinal and

particularly post-transition data to evaluate other outcomes of their use in glomerular disease or any kidney disease [24]. A group in Japan utilized the 14-item TRANSITION-Q to assess the efficacy of their transitional support clinic by measuring changes in transition readiness. They found an increase in scores from the intervention but did not follow the patients post-transition to assess the success of transition [47]. A Chinese group used the STARx questionnaire and found an increase in transition readiness by implementing a nurse-led video-based education program particularly for younger adolescents [44].

Conclusion

Overall, the transition from pediatric to adult care in patients with glomerular disease is a complex and underappreciated process which deserves further attention and investment from health care staff and patients alike. It requires a degree of creativity and flexibility given the widely varied access to resources including dedicated pediatric or adult providers in nephrology and often rheumatology. Support staff such as in social work, nursing, psychology, and/or case management also promote a successful transition. These team members can assess readiness, help patients develop self-care skills, identify gaps in resources, provide a plan for a coordinated transfer, and follow up on whether the transfer successfully occurred. The involvement of social work or case management may be particularly important given the documented influence of socioeconomic status and insurance status on disease and transition outcomes.

Successful transition is important for slowing loss of kidney function, maintaining physical and psychosocial function, and possibly even for managing health care utilization and costs. More studies are needed to form definite conclusions as to whether improved HCT could prevent or delay morbid, though also expensive, events including the need for dialysis or transplantation. There are recommended frameworks, guidelines, and toolkits available to provide a roadmap for potential success, but they have to be applied to each particular local environment with its own challenges. Not every clinic has access to the full range of support team members, and despite guideline suggestions, a combined clinic is not always feasible. Despite this variability in resources, at least some aspects of transition interventions which have proven successful could be applied across more settings. Investment in transition with a fully supported team could improve outcomes for young adults, which impacts their health and well-being for the rest of their lives.

Additional research is needed in which strategies and interventions promote a successful transition and transfer for patients with glomerular disease. Our review of the currently available data in HCT in glomerular disease illuminates several specific areas which need further study. We propose several research priorities in the future. First, there has to be more clearly established outcomes to measure. Important outcomes to assess the success of transition may include process-related data, such as rates of transfer to adult care and proof of establishment of continuity of care beyond the first visit. Other important outcomes to measure are disease-related metrics across the transition period, such as measurements of disease activity and kidney function, for example, the slope of glomerular filtration rate or proteinuria. Next, we propose that outcomes should be studied with more interventions; in many places, creative solutions are being applied, possibly with success. Without studying and then reporting outcomes, however, successful interventions may not be realized or applied elsewhere. Studying health care cost could also identify areas of cost savings related to successful HCT, which could further promote investment into HCT. Implementing a successful transition program does take planning and effort, but it is an important facet of young adults' health in glomerular disease. Ultimately, to improve outcomes including morbidity, mortality, and patient and provider satisfaction, it is necessary to dedicate attention and resources to HCT in AYA patients with glomerular disease.

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Conflict of Interest Statement

The authors have no conflicts of interest to declare.

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Author Contributions

All authors contributed to article concept, literature review, writing, and editing. J.S.C.: conceptualization, project administration, writing, and editing. M.K. and J.G.: conceptualization, writing, and editing. J.A.M.: conceptualization, writing, editing, supervision.

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